ISL-backend -- CVIP 2026 Research Upgrade

Summary of changes made in this engagement. Generated 2026-08-12.

What this engagement was

You asked for the ISL-backend project to be taken from its current state to the strongest scientifically defensible version possible ahead of a CVIP 2026 research paper -- audit everything first, fix what needed fixing, and never fabricate a result. No paper text was written; this was entirely about making the project itself honest, reproducible, and correct.

What was found

- The dataset had grown to 31,688 samples across 31 of 35 target classes (26 letters + 5 of 9 word-gestures) since the last time results were computed -- but the results file, both data-collection docs, and the paper draft all still described an old 842-row, 18-class snapshot from before a large Kaggle dataset was merged in.
- 97% of the current dataset comes from a Kaggle image dataset (CC BY-SA 4.0 license, now properly attributed in LICENSE); the remaining 3% is webcam recordings you made yourself. No real multi-signer data exists anywhere in the project.
- The single most important finding: the system's apparent 99.9% accuracy is almost entirely a same-source artifact. Trained on the Kaggle images and tested on your webcam recordings (or vice versa), accuracy collapses to 29-62%. This cross-source gap, not raw accuracy, is the project's real scientific finding.
- That gap is not explained by having too little webcam data (tested and ruled out), and is only partly explained by hand-orientation/mirroring differences between the two recording setups. A targeted fix (treating one-handed signs the same way regardless of which hand performed them) measurably closes part of the gap -- proven with statistics, not just a lucky run -- but does not fully explain it. Some letters (A, E, F, K) still fail completely and remain an open question.
- Manually inspecting the original Kaggle images (outside this repo) turned up something not previously documented: most images are tiny 128x128 pixel close-up crops, four letters have extra higher-resolution images mixed in from a different source, and this resolution gap versus your webcam's HD video is a likely (but not yet proven) additional cause of the remaining failures.

What was fixed

- LICENSE: added attribution and share-alike terms for the Kaggle-derived data.
- evaluate.py rebuilt end-to-end for the current dataset: duplicate rows removed before training, cross-source evaluation added, statistical confidence intervals added, a neural-network baseline added alongside the existing Random Forest/SVM/kNN comparison.
- train_model.py deduplicated, and the shipped model retrained on clean data.
- docs/data_collection_protocol.md and docs/reproducibility_checklist.md rewritten to match the real current dataset, including a clear statement that no signer-independence claim can be made, and confirmation that all existing webcam recordings are your own.
- Two new analysis scripts (analyze_domain_shift.py, build_invariant_features.py) added for the cross-source investigation and its candidate fix, plus a figure-generation script producing 8 new charts, all built only from real computed numbers.
- The whole project was then reviewed three separate times by independent, deliberately hostile reviewers (a computer-vision reviewer, a machine-learning reviewer, and a sign-language/ethics reviewer) to find anything that would get a paper rejected. Their genuinely fixable findings -- missing statistics, a flawed comparison, a missing baseline -- were fixed; everything else is written down honestly as a limitation.

What was deliberately NOT changed

- paper/main.tex was left untouched. It currently argues a finding (accuracy collapses under a stricter data split) that your own current results now contradict -- it needs a new structure centered on the cross-source finding, not a numbers update. Writing that new paper is the next step, and was intentionally kept out of scope for this engagement.
- The redundant isl_dataset_kaggle.csv file was kept (not deleted) as a provenance record of exactly what was imported -- it was already excluded from git before this engagement.
- MediaPipe detection-confidence settings differ slightly between the live app and the offline import scripts. This is now clearly documented in the code and in the reports, but was not changed, since altering it retroactively can't fix data already collected.

Where to look for full detail

- PROJECT_DOSSIER_FINAL.md -- the complete research dossier: every finding, what the paper can and cannot claim, and a recommended new paper structure.
- RESULTS_FINAL.json -- the same findings as structured data, each one tagged as confirmed, manually-observed, hypothesis, or future work.
- results/RESULTS.md and the 8 new charts in results/ -- the underlying numbers and figures.

This PDF is a plain-language summary for your own record. Login credentials and any secrets are not included here or anywhere in the repository.